

Dissertation report

by Qibang Chen

Submission date: 03-Sep-2022 08:14PM (UTC+0100)

Submission ID: 185666120

File name: Dissertation_report.pdf (2.1M)

Word count: 6884

Character count: 37623

Simulation of heart movement in a cardiac cycle

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A dissertation submitted to The University of Manchester for the degree of
Master of Science in Advanced Control and Systems Engineering
in the Faculty of Science and Engineering

Year of submission

2022

Student ID

10852616

Author; Qibang Chen

36

Supervisor: Dr Fumie Costen

School of Engineering

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Abstract

Diagnosis of dyspnea is currently performed by measuring the size of the lungs through MRI scanners, which is expensive and time-consuming for normal patients.[1] The wider research team behind this project is developing a device that estimates the size of the lung by sending microwaves through the torso and thus detecting anomalies in the lung structure. Considering ethical and medical concerns, a digital human phantom is required for the testing and evaluation of the device.

This project basis on the developed digital phantom that replicates respiratory movements. The model to be modified will be valuable for the testing and evaluation of the microwave device, providing the variance of heart shapes and volumes during a cardiac cycle. The first attempt to change the shape of the heart was made by Jang and Yifan. This project identifies the merit of each method and provides a code to change the shape of the heart by combining their merits and improving the accuracy of the model.

The main objective of this project is to develop a model that simulates the heart movement during the cardiac cycle. The main algorithm consists of dividing the heart and compressing the heart. The division of the heart is implemented by dividing each transverse plane into four parts - left atrium, left ventricle, right atrium, and right ventricle by two lines. The compression of the heart is implemented by modifying the value of the voxel in the 3D data set, and outputs are generated in the form of slices through the phantom. An evaluation is performed to assess the validity and accuracy of the model, by comparing the outputs to real-life MRI data.

Acknowledgements

I cannot imagine what work has been done to be so far in this project and I am honored to participate in the project. Firstly, I ¹thank my supervisor, Dr. Fumie Costen for her guidance and support on the overall project. Then, I thank my GTA Qiyuan for his patience and support in programming. Besides, I would like to thank Yifan, Jang, and other co-workers for their contribution to the project. Thank you.

1 Introduction

1.1 Background and motivation

Dyspnea (commonly known as shortness of breath) is a common medical condition affecting about 50% of patients admitted to acute, tertiary care hospitals and about 30% of ambulatory patients. [2] It can arise from many different underlying conditions and is sometimes a manifestation of a life-threatening disease. Thus, accurate and convenient evaluation and targeted diagnostic studies are of vital significance. [1]

Respiratory Dyspnea is the sensation of difficult or uncomfortable breathing. The patient cannot exhale to a maximum degree while suffering from dyspnea. It is a subjective experience perceived and reported by an affected patient. The condition can be improved if the patients notice their breath. Before this subjective approach, patients need to detect that they are not exhaling to maximum capacity. [3] This detection can be performed by measuring the size of the lung during respiration, which is currently performed by modern medical devices such as MRI. While the utilization of MRI provides accurate diagnosis and results, the MRI scan and the segmentation are too expensive and not convenient to perform on each patient, considering the prevalence of dyspnea. For instance, a single MRI scan costs the NHS 130 pounds and more than one hour. [4]

The wider research team behind this project guided by Dr. Fumie is trying to develop a device that estimates the size of the lung by sending the microwave to the torso and analyzing the signals received on the other side of the torso. As the development of this kind of device cannot be performed using real and alive human, we need to produce a wide range of digital human phantoms which has a variety of lung size to imitate respiration.

The work so far handled the reduction of the volume of the lung in a two-dimensional way, i.e., the process was carried out plane by plane, which is handled by Dale. The movement of the heart and diaphragm due to the breathing was handled by Tomos. This does not produce a movie but produces one instance after the exhalation. This also means that the shape of the heart was not changed during the exhalation. However, the heart is relaxed and contracted more than once during one cycle of breathing which includes inhalation and exhalation. To produce a more realistic movie of the movement of the lung we should take into account the change of the heart shape during the one cycle of breathing. Furthermore, the movement of the ribcages (bones

around the lung) is untouched. The first attempt to change the shape of the heart was made by Jang and Yifan, at the same time, separately.[5]

This project understands and estimates the merits of Yifan and Jang's work. The basis on these previous works provided by Dr. Fumie, a more accurate algorithm for modeling heart compressing process is developed.[6]

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1.2 Aims and objectives

This project contributes to the overall development of the DHP, making the model more realistic and accurate. The main aim is to develop a model that simulates the movement of the heart in a cardiac cycle, reflecting the real heartbeat motion. The model will utilize and modify work on previous progress and thus ensuring a continuation of the overall DHP development.

This project first understands and estimates the approaches taken by Jang and Yifan and then identify the merits of each method. A modified algorithm of dividing and compressing the heart is developed by taking these merits. This project increases the accuracy of both heart division and heart compressing processes.

Objectives:

- Analysis and combining merits of previous works done by Yifan and Jang.
- Design a modified algorithm to realistically change the shape of the heart.
- Design a method to evaluate the performance and accuracy of the outcome

1.3 Report structure

This report is developed in 6 chapters.

Chapter 1 -Introduction. Introduces the reader to the context and background of this project. Sets the expectations for the rest of the paper.

Chapter 2 -Literature research. Covers the anatomy and physiology of the relevant body parts. Respiration and cardiac cycles are reviewed and examined, necessary to comprehend and appreciate the decisions made in the development of the project.

Chapter 3 -Technical background. Covers the concepts and technologies necessary to follow the technical work performed on this project.

¹Chapter 4 -Method. Provides a detailed description of the methods, algorithms, and designs used in the development.

Chapter 5 -Results and discussion. Analyses the outcome and performance of the program. Presents inaccuracies and complications alongside their possible improvements and solutions.

Chapter 6 -Conclusion and future work. ¹Reflects on the achievements of the project. Expands on plans for continued work for the overall development.

2 Literature review

2.1 Directional terms

Standard anatomical positions are the basis of a clear and consistent description of human anatomy and physiology. ¹ Standard references of the orientations in the human body are used throughout this report. The ¹ directional terms are visualized and defined in the table presented in ¹² Figure 2.1. There are three planes most commonly used in anatomical and medical imaging the sagittal, frontal, and transverse planes. [7]

- ⁶ ● The frontal plane is the plane that divides the body or an organ into an anterior (front) portion and a posterior (rear) portion. The frontal plane is often referred to as a coronal plane. ("Corona" is Latin for "crown.")
- The sagittal plane is the plane that divides the body or an organ vertically into the right and left sides. If this vertical plane runs directly down the middle of the body, it is called the midsagittal or median plane. If it divides the body into unequal right and left sides, it is called a parasagittal plane or less commonly a longitudinal section.
- ¹² ● The transverse plane is the plane that divides the body or organ horizontally into upper and lower portions. Transverse planes produce images referred to as cross sections.

In this project, all relative orientation terms will corresponding to the figure below. The coronal plane refers to the slice dividing the body into front and back. The sagittal plane refers to a slice dividing the body into right and left. The transverse plane refers to a slice dividing the body into top and bottom.

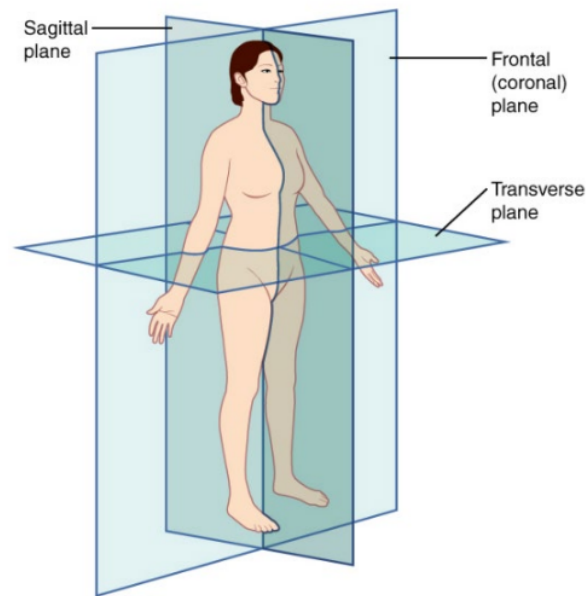


Fig. 2.1. Three commonly used planes of the body.[7]

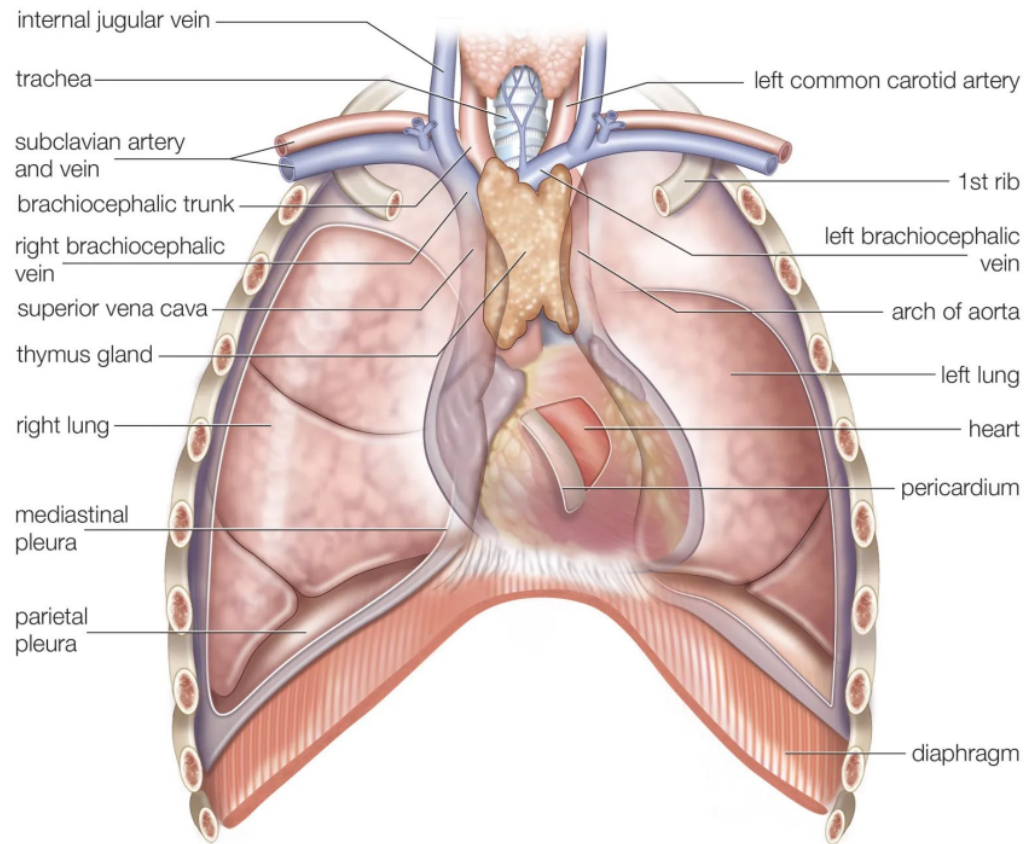
2.2 Human heart structure

The ²³heart is the organ that pumps blood, circulating it to all parts of the body. ¹Blood circulation supplies body tissues and organs with oxygen and nutrients while transporting metabolic waste products such as carbon dioxide back to the lungs. The heart is located between the lungs within the second and sixth rib. In a human, the size of the lung is about the size of a clenched fist and weighs between 300g to 350g in adult males and 250g to 300g in the adult female. ¹¹The human heart is a four-chambered double pump with its right and left sides fully separated by a septum and subdivided on both sides into an atrium above and a ventricle below.[8]

2.2.1 Location of heart

As shown in Fig 2.2, in normal ³humans, the heart is situated between the two lungs and slightly to the left of the central line of the human body, ³behind the breastbone. It rests on the diaphragm, the muscular partition between the chest and the abdominal cavity. [8]

Heart in situ



© 2010 Encyclopædia Britannica, Inc.

Fig. 2.2. Heart in situ [8]

2.2.2 Heart wall

³ The heart consists of several layers of a tough muscular wall, the myocardium. A thin layer of tissue, the pericardium, covers the outside, and another layer, the endocardium, lines the inside.

[8]

⁸ There is a transverse coronal sulcus near the bottom of the heart, which circles the heart and is the symbol that separates the atrium and the ventricle outside the heart. There are anterior and posterior interventricular grooves on the front and back of the heart, which demarcate the surfaces of the left and right ventricles.

2.2.3 Interior structure

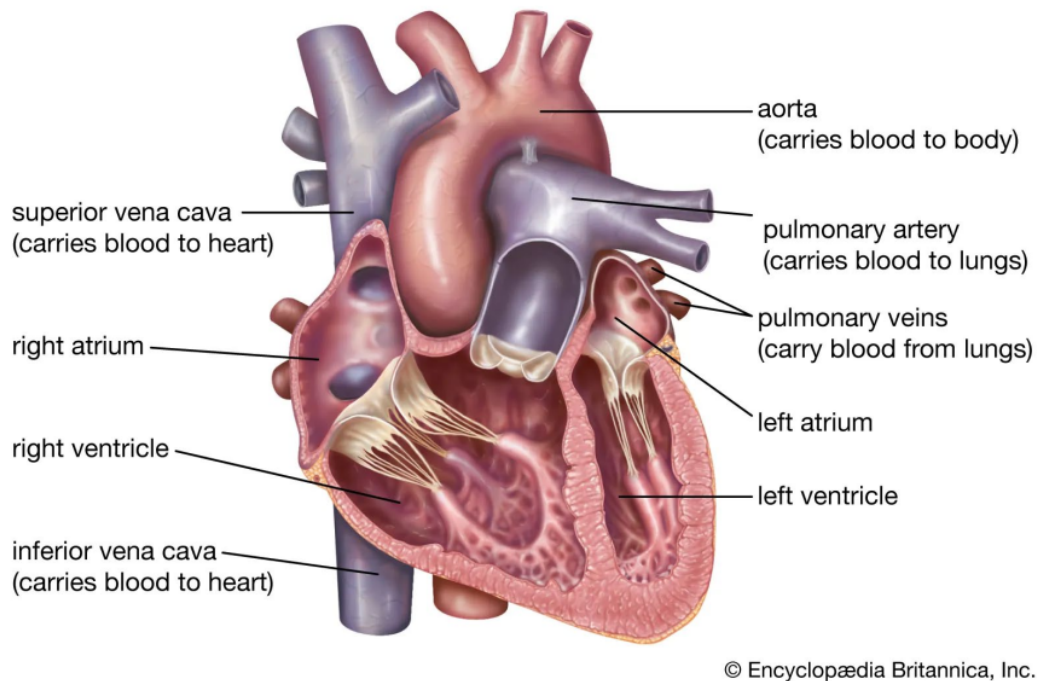


Fig. 2.3. Gross anatomy of a heart. [8]

The structural components of the human heart ¹ can be identified from an interior view of the heart as shown in figure 2.3. ¹ At the superior end of the heart, the great veins and arteries are located, which are responsible for carrying blood into and out of the heart. At the inferior of the heart, ³⁸ the heart consists of four chambers and their respective valves. [9]

² The heart is a single organ that can be considered as two pumps that propel blood circulation in two individual circuits. The right atrium receives venous blood from the head, chest, and arms via the large vein called the superior vena cava and receives blood from the abdomen, pelvic region, and legs via the inferior vena cava. Blood then passes through the tricuspid valve to the right ventricle, which propels it through the pulmonary artery to the lungs. In the lungs, venous blood comes in contact with inhaled air, picks up oxygen, and loses carbon dioxide. Oxygenated blood is returned to the left atrium through the pulmonary veins. Valves in the heart allow blood to flow in one direction only and help maintain the pressure required to pump the blood. [8]

Before the literature research on the cardiac cycle, which is the main part of the simulation, it should be highlighted that the heart structure is not symmetrical. The left ventricle is significantly larger than the right ventricle, which causes the heart tilts left slightly. Therefore, a mid-sagittal plane will not cut the heart into the equal right ventricle and left ventricle. Instead, each coronal plane is divided into the left atrium, left ventricle, right atrium, and left ventricle by two sloping lines.

2.3 Cardiac cycle

The cardiac cycle is defined as the performance of the human heart in the process of a heartbeat. By combining the organization of the heart and the mechanisms of a cardiac cycle, the structural derivation of the human heart during a cardiac cycle can be understood. A cardiac cycle consists of two periods, namely (ventricle)diastole and (ventricle)systole.

In a cardiac cycle, ventricular diastole is ahead of ventricular systole coordinating with atrial systole ahead of atrial diastole. If the average heart rate of adults is 75 beats per minute, each cardiac cycle is 0.8 seconds on average [10]

2.3.1 Ventricle Diastole

Diastole refers to the diastole of the ventricle here, which occurs during the first half of the cardiac cycle. The organization of the heart muscle will be relaxed. Diastole causes the filling of the chambers with blood, averaging 0.45 to 0.55 seconds.[12]

As shown in figure 2.6, during diastole, the internal pressure decreases, and blood from the vena cava returns to the heart. The atria and ventricles are relaxed in the ventricular diastole period, and the atrioventricular valves are open. Oxygen-depleted blood returning to the heart from the body following the last cardiac cycle passes through the superior and inferior vena cava and flows to the right atrium. [13]

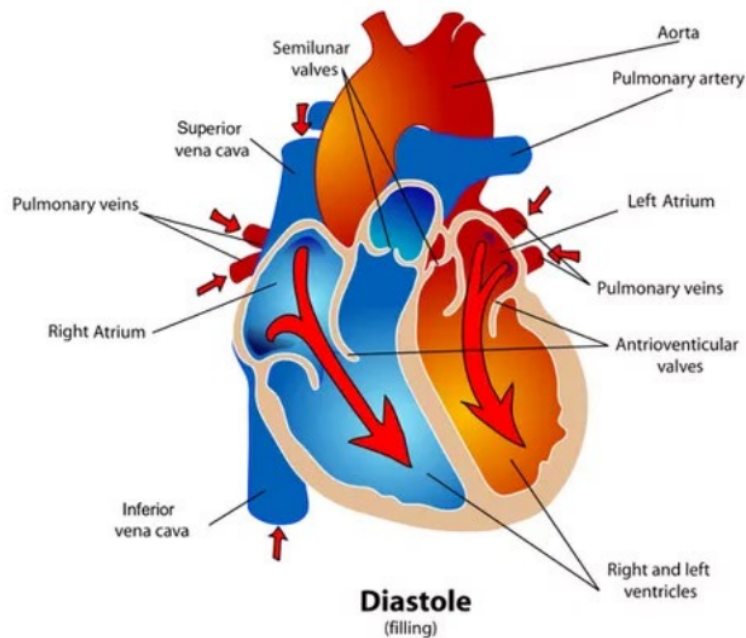


Fig. 2.4 ventricle diastole period of the cardiac cycle [13]

2.3.2 Ventricle Systole

³ Systole refers to the systole of the ventricles, at the same time the atrium is in the systole period, which occurs during the latter half of the cardiac cycle. Systole causes the ejection of blood into the aorta and pulmonary trunk. Lasting usually 0.25 to 0.35 seconds. [12]

As shown in figure 2.5, during the ventricle systole period, the internal pressure rises, pumping blood into the arteries. The ⁵ ventricular contraction causes oxygen-depleted blood from the right ventricle to be pumped to the pulmonary artery. The pulmonary valve prevents blood from flowing back into the right ventricle. The pulmonary artery carries de-oxygenated blood along the pulmonary circuit to the lungs, where the blood circulation collects oxygen and returns to the left atrium of the heart through the pulmonary veins. [13]

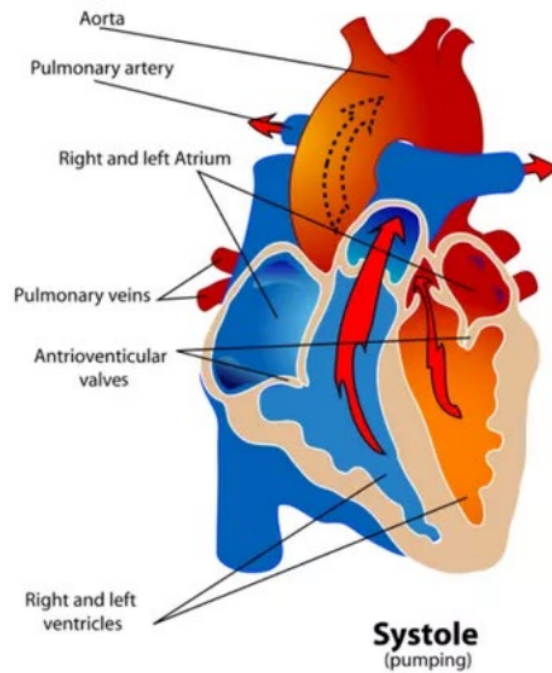


Fig. 2.5 ventricle systole period of cardiac cycle [13]

2.3.3 Cardiac cycle phases

As shown in figure 2.6, the cycle diagram displays the heart movement during the cardiac cycle. It can be detected that the atria diastole period is not the same as the ventricle diastole period. Besides, the atria systole period is not the same as the ventricle systole period. In each cardiac cycle, the two atria contract first, with the right atrium contracting slightly ahead of the left atrium. The two ³⁵ventricles contract after the atria begin to relax, and the left ventricle contracts slightly before the ¹right ventricle. At the end of ventricular diastole, the atria begin to contract again. ¹Atrial systole averaged 0.11 seconds and diastole averaged 0.69 seconds. Ventricular systole averaged 0.27 seconds and diastole averaged 0.53 seconds.[14]

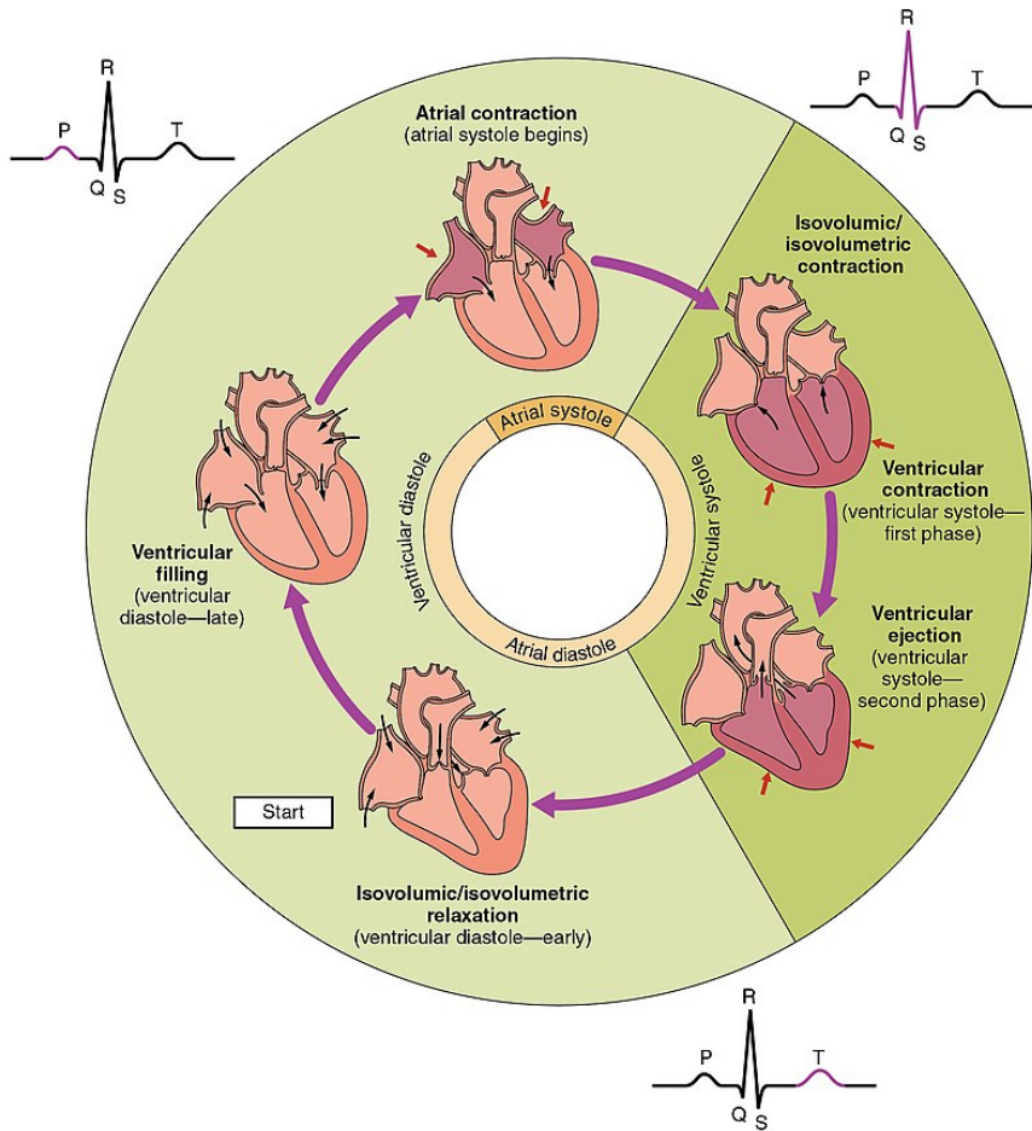


Fig. 2.6 pressure and volume of the ventricle[11]

2.4 Structural change of heart in a cardiac cycle

The cardiac cycle is indeed a sequence of electrical and mechanical activity during the period of systole and diastole. As shown in figure 2.7, the variance of ventricle pressure and ventricle volume as a function of time series can be detected, which will be the basis of the heart

movement model. [15] The cycle of both volume and pressure is similar for the right and left ventricles. The volume will decrease from about 130ml to 55ml during the systole period and increase from 55ml to 130ml during the diastole period. When the ventricle is fully compressed, it will reduce to 45% of its size in relaxation.

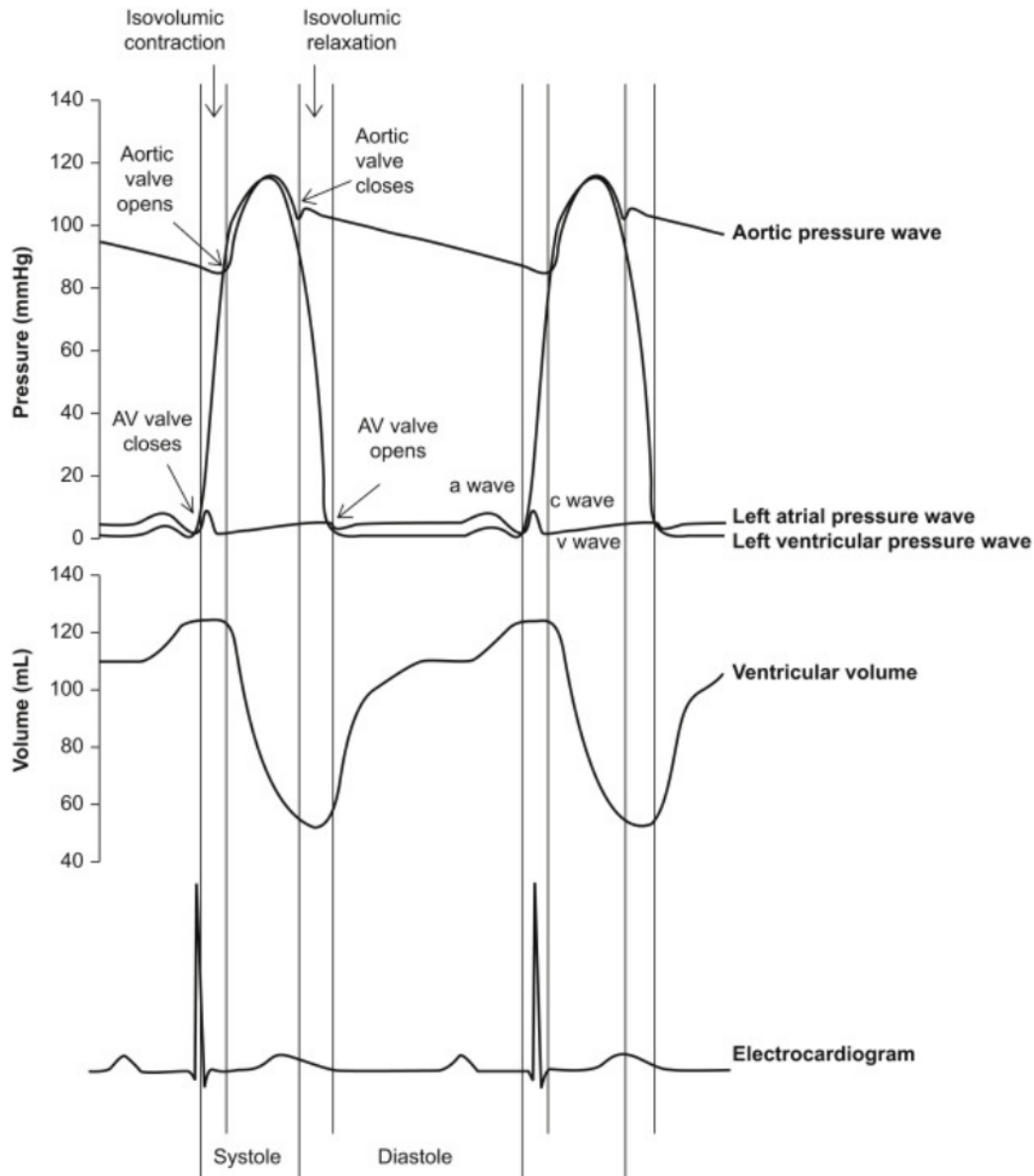


Fig. 2.7 Wiggers diagram showing the pressure and volume of the ventricle.[15]

3 Technical Background

3.1 Digital Human Phantoms

Compared to researching and operating on live creatures, utilizing digital human phantoms can be repeatable and more convenient. Relative ethical problems can be avoided once the computational model is developed.

The widespread availability of cutting-edge computing devices has been promoting the development of computational anthropomorphic models of the human anatomy for medical imaging modalities and dosimetry calculations. Nowadays, more accurate models of the human body with three or more dimensions are making the simulation of medical imaging more accurate. [16]

There are mainly three types of digital human phantoms, which can be defined as:

Mathematical model: The model is constructed by equations and functions. It is obtained through the analysis of the physicochemical biological properties.

Digital model: The model consists of voxel values. It is mostly obtained by scanning real objects through the technology of CT and MRI, etc.

Hybrid model: It is a combination of a mathematical model and a digital model.

In this project, as in most cases, the developed heart movement model is a hybrid model. The obtained MRI data is collected and stored in a 3D array, which makes the digital model accurate in the first place. Thus, the static original static model can be referred to as a digital model. With the mathematical functions of dividing and compressing the heart, the static model is becoming more flexible with less computation process. With this method, the overall hybrid heart movement model is developed.

3.2 Portable Grey Map (PGM)

The PGM format of the image is simple and raster. It is convenient for users and developers to manipulate on and converse. A PGM file contains as many values as pixels of the image, each can be referred to as the intensity or brightness of the corresponding pixel in the image. A larger value corresponds to a whiter pixel, and vice versa.

The specific PGM format utilized in this project consists of the following parts:

1, Specific PGM file format type (In this project, "P2" is used).

2, Notes and comments.

3, ¹Width of the image, height of the image.

4, The maximum grey value, in ASCII decimal

5, Pixel values of the image.

```
1 P2
2 # Shows the word 'Heart' as an example of PFM file format used in this project
3 29 7
4 10
5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
6 0 9 0 0 9 0 0 7 7 7 7 0 0 0 5 0 0 0 3 3 3 0 0 9 9 9 9 9 0
7 0 9 0 0 9 0 0 7 0 0 0 0 0 5 0 5 0 0 3 0 3 0 0 0 0 7 0 0 0
8 0 9 9 9 9 0 0 7 7 7 7 0 0 5 5 5 0 0 3 3 0 0 0 0 0 5 0 0 0
9 0 9 0 0 9 0 0 7 0 0 0 0 0 5 0 5 0 0 3 0 3 0 0 0 0 3 0 0 0
10 0 9 0 0 9 0 0 7 7 7 7 0 0 5 0 5 0 0 3 0 3 0 0 0 0 1 0 0 0
11 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
12
```

Fig. 3.1 An example of PGM file structure created by the author.



Fig. 3.2 The representing image corresponding to the PGM file in Figure 3.1

¹3.3 Data Provided by RIKEN

The digital human phantom used in this project was provided by RIKEN (A high-performance computer used by the team behind this project). The entire project follows the University of Manchester's Code of Research Ethics.

The DHP¹ is produced by scanning a healthy normal Japanese man aged around 55. One slice was scanned every millimeter²⁶ from the top of the head to the bottom of the feet. 600 slices of 1mm*1mm voxel images of the transverse plane are selected from the overall DHP.

Each image is named regularly from v1_00200.pgm to v1_00799.pgm. Each DHP image corresponds to a one-millimeter depth on Z-axis. Each slice is viewed as a 2D image of 265(X-axis) *490(Y-axis) pixels. For example, v1_00340.pgm refers to the 340th transverse plane counted from the top of the head. as shown in figure 3.3.

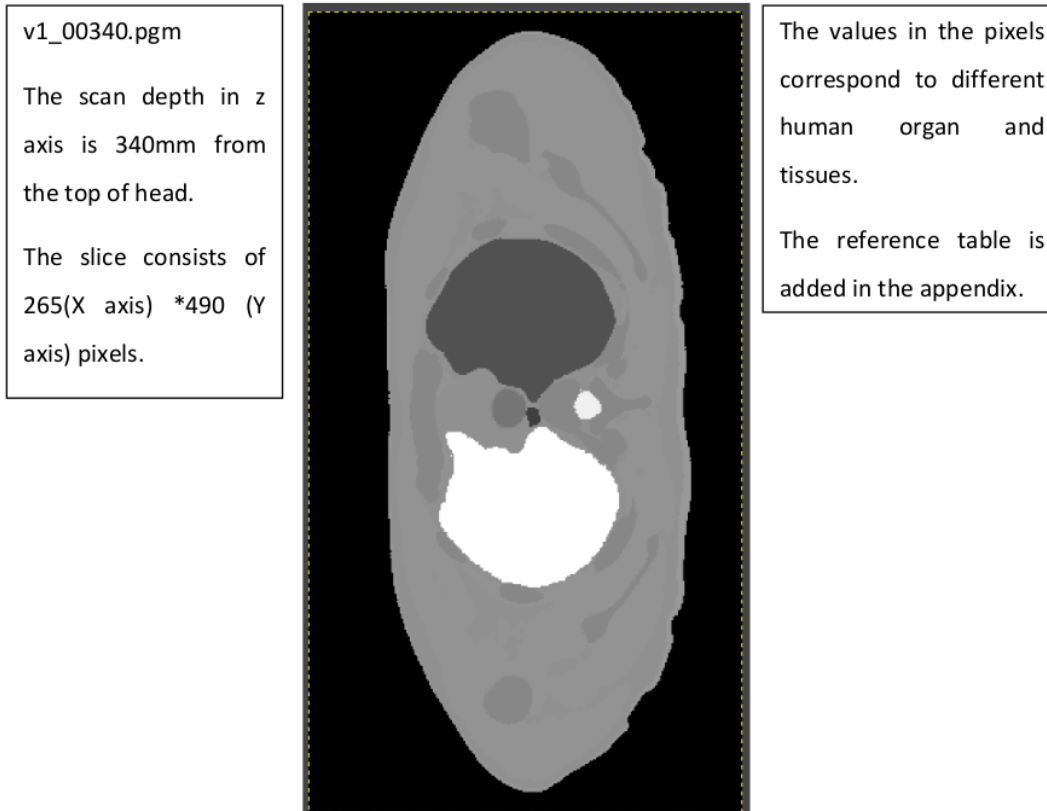


Fig. 3.3 v1_00340 of DHP provided by RIKEN

In this report, the sequence of three orientations of planes are defined as: Transverse plane - Z-axis position; Coronal plane - X-axis position; sagittal plane - Y-axis position. The actual length can be calculated by multiplying two axes (the unit is 10^{-6} m^2) while the actual volume can be calculated by multiplying three axes (the unit is 10^{-9} m^3).

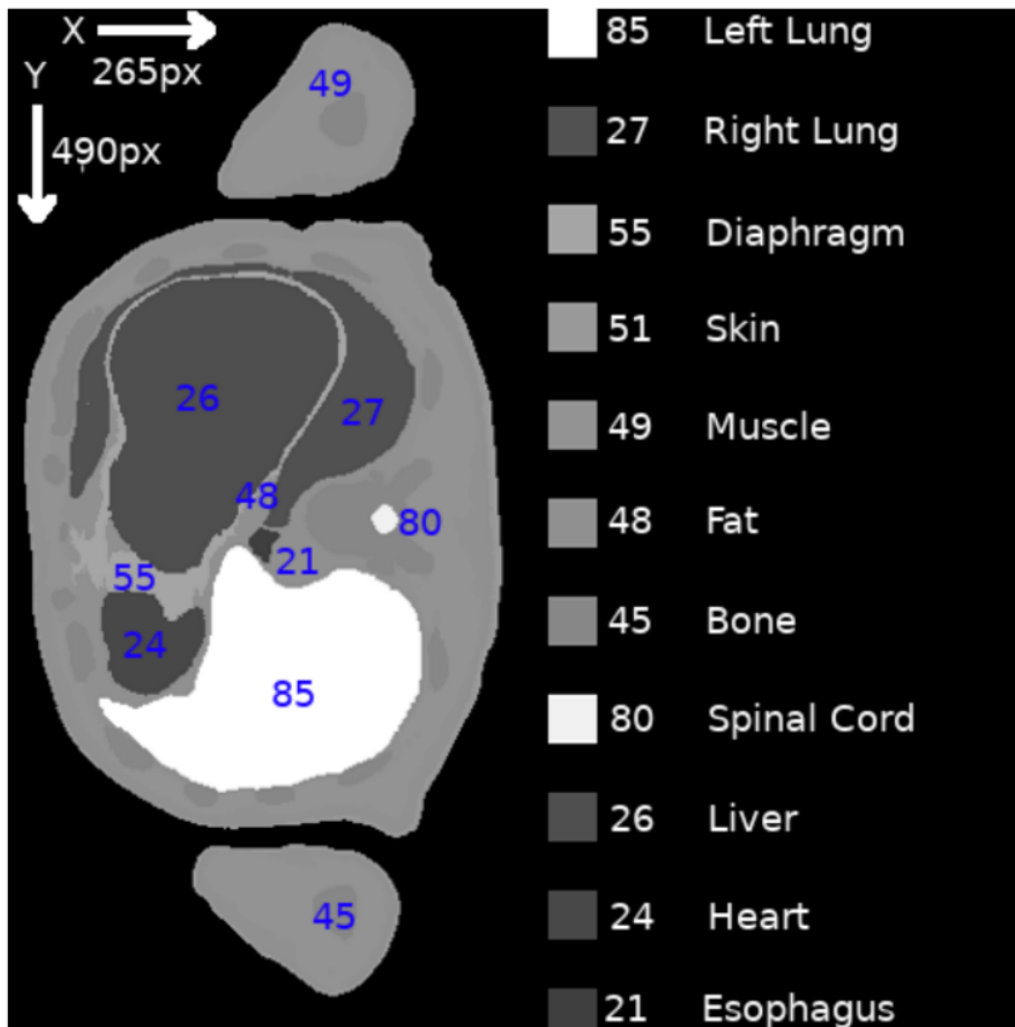


Fig. 3.42 The PGM format file provided by RIKEN with different organs labelled with corresponding greyscale values.[17]

4 Methods

4.1 Overview of the project

The following sections introduce the structure and algorithms of the project. The block diagram as shown in Figure 4.1 outlines the process from inputting data to outputting results. The programming language used for this project is FORTRAN 90, which is also used by previous work and the team behind this project. The program is written in VSCODE for Windows and compiled with the MinGW-64 compiler.

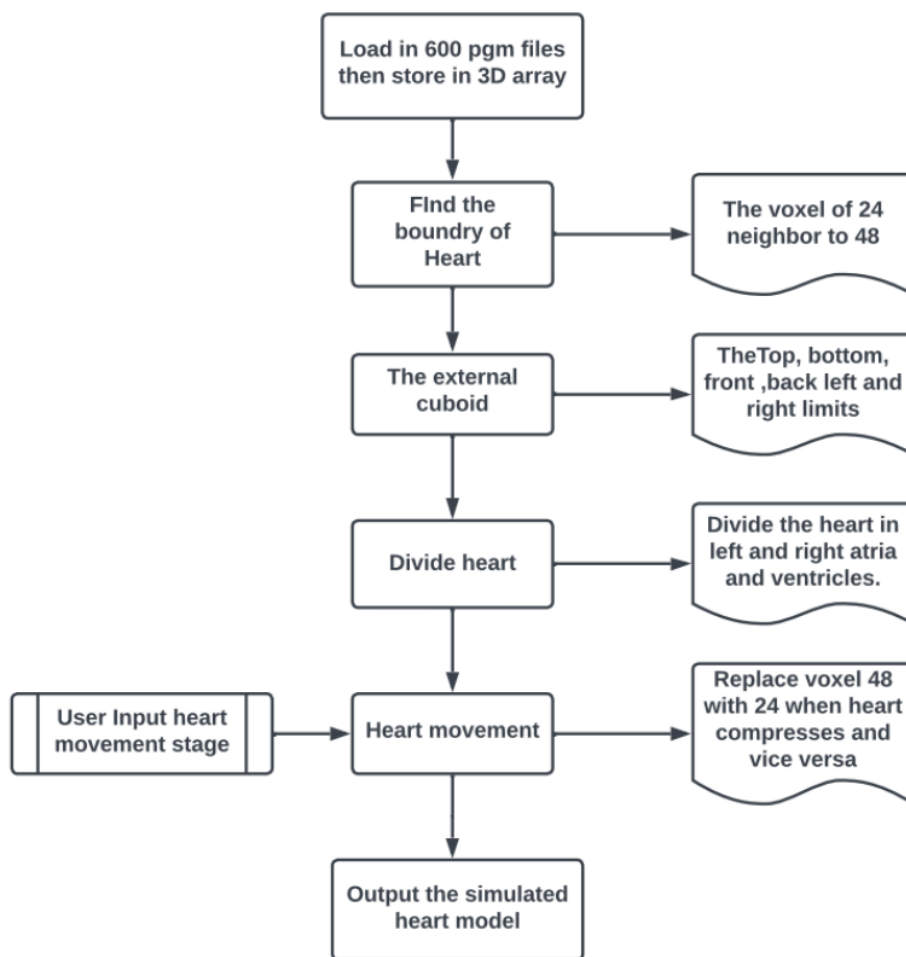


Fig. 4.1 Main structure of the heart movement simulation created by the author.

The overview of the project consists of 6 steps, each step is illustrated in detail in the following sections.

1. The data consists of 600 PGM format files provided by RIKEN and is loaded by the .exe file. These 2D images are stored in a 3D array as a digital simulation of the human body.
2. The boundary of the heart is identified by finding the voxel value 48, which refers to heart, that is neighboured to yvoxel value 24, which refers to fat.
3. An external cuboid of heart is found by combining the left, right, upper, bottom, front, and back limits of the heart.
4. The ²⁸heart is divided into left atrium, right atrium, left ventricle, and right ventricle in each coronal slices.
5. The ventricle systole process is simulated by replacing the voxel value 48 with voxel value 24. In the same way, the ventricle diastole process is simulated by replacing voxel value 24 with voxel value 48.
6. According to the user input, including the stage of the cardiac cycle and the orientation of output files, the outputs are generated and stored.

4.2 Load in data

The data consists of 600 PGM files are provided by RIKEN. The first step of the project is to load this data and then store these 2D images into a 3D array as a simulation of part of the human body.

The initial images in PGM format can be viewed as 600 slices of the transverse plane of human body. These images are representing a millimeter height slice of the human body. This project uses index i to point to the corresponding depth of the slice. Each image consists of 265(X axis) *490 (Y axis) pixels. The order of pixels in X-axis is pointed to by index j and the order of pixels in Y-axis is pointed to by index k . Therefore, every single voxel of the DHP representing 1mm*1mm*1mm*1mm cube of the human body is named and can be called as:

```
data_input_int(i,j,k) = 48
```

The purple point in figure 4.2 is a representation of the value and numbering of each voxel in the DHP. The location of each 1mm*1mm*1mm*1mm cube of the human body is represented by: The digital human phantom is constructed once the data is loaded and the 3D aris ray constructed. As an illustration, the slice of sagittal plane with index $j=302$ is presented in Figure 4.2, each line in

the sagittal plane from index $i = 200$ to $i = 799$ is corresponding to a transverse plane. For example, the V_00471.pgm file with index $i = 471$ is presented in Figure 4.3.

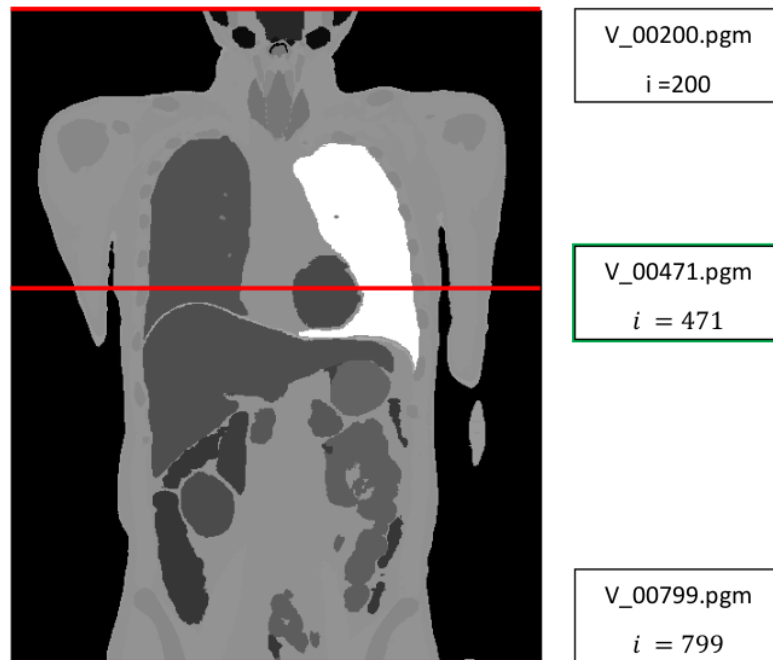


Fig. 4.2 The original coronal plan with index $j=302$

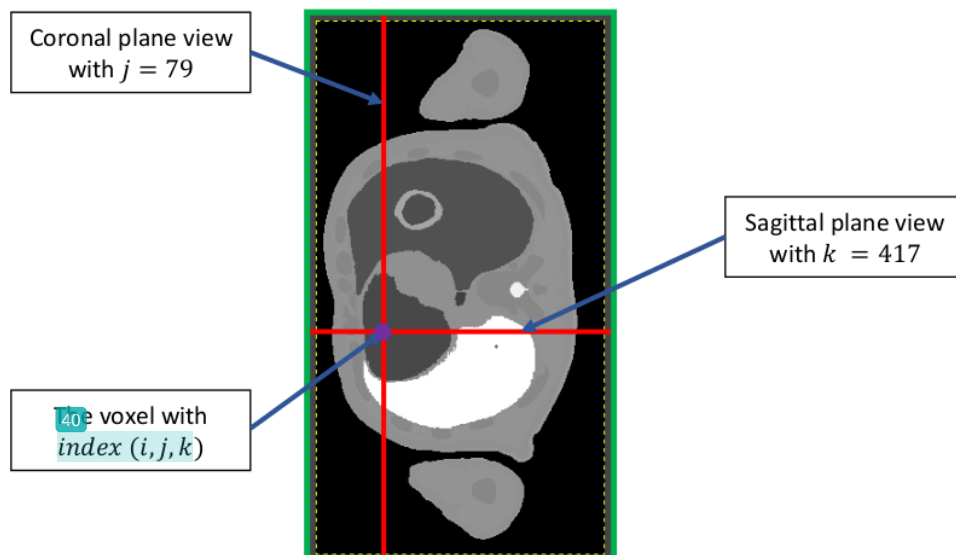


Fig. 4.3 The original transverse plan with index $i = 471$

4.3 Algorithm of dividing the heart

4.3.1 The external boundary of heart

When the data is loaded, the heart is currently viewed as a whole with a voxel value 48. The borders of the heart are found by traverse scanning and detecting cardiac voxel values (48) adjacent to fat (24). The limits of the Z-axis, which represents the top and bottom of the heart, are found by traversing through all coronal planes. The limits of the X-axis, which represents the right limit and left limit of the heart, is found by traversing through all transverse planes. The limits of the Y-axis, which represents the front and back of the heart, are found by traversing through all sagittal planes. For both transverse view, coronal view, and sagittal view, the method is the same.

An external cuboid can be constructed with these six limits. The length, width, and height of this circumscribed cuboid are defined as the length, width, and height of the heart. For each coronal slice, a circumscribed rectangle can be defined. For example, as shown in Figure 4.4, four red lines represent the limit of the right, left, top and bottom boundary. The green triangle is the circumscribed rectangle for the coronal slice with index $k = 295$.

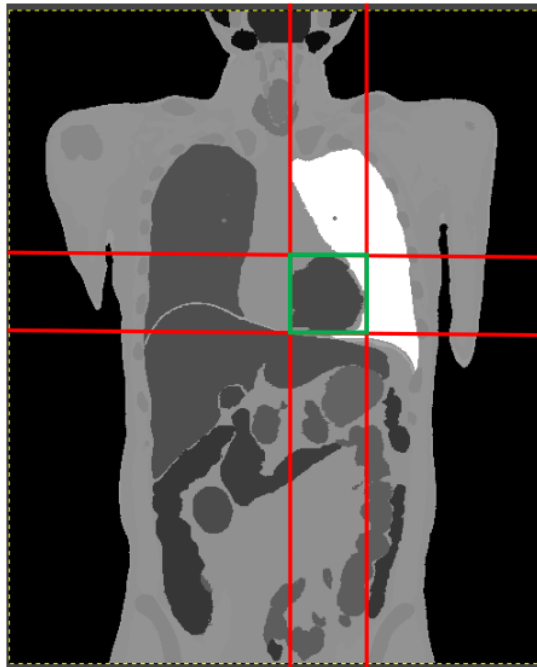


Fig. 4.4 The original coronal plan with index $j=295$

4.3.2 Division of heart

The division of the heart is to be made in the transverse planes, the limits of left, right, top and bottom are stored in a series of arrays: Heart_boundry_k (Left_Limit, Right_Limit, Front_Limit, Back_Limit). According to the literature research part, the midpoint of the heart is defined as the voxel located at 33.33% of the length and 33.33% of the width of the circumscribed rectangle for each transvrse slices. The midpoint is calculated by:

```
midpoint_k = left + 0.6666*(right - left)
midpoint_j = top + 0.3333*(bottom - top)
```

As shown in figure 4.5, to illustrate the midpoint and circumscribed triangle for each transverse plane, the transverse plane with index $i = 477$ is presented. In this specific transverse plane, two yellow lines are the left and right limits while two green lines are the front and back limits. The red point is the defined midpoint, here the midpoint is located by index $(i, j, k) = (477, 101, 264)$.

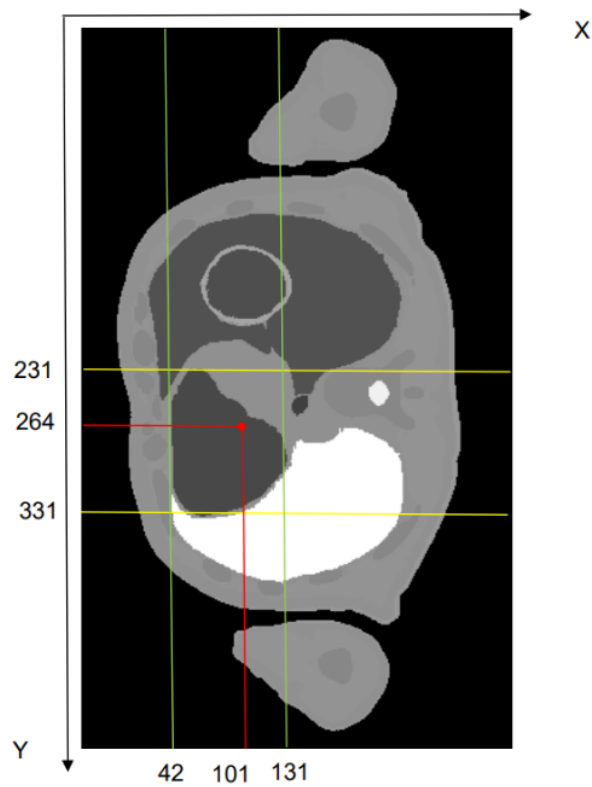


Fig. 4.5 The original transverse plan with index $i = 477$

After determining the midpoint of each transverse slice, two-line segments are used to divide the heart into four parts: ¹⁵ left atrium, left ventricle, right atrium, and right ventricle. The equations of ¹⁵ the two lines can be written as:

$$\begin{aligned} n &= -1 * \text{midpoint_j} - 1.732 * \text{midpoint_k} \\ m &= -1 * \text{midpoint_j} + 0.577 * \text{midpoint_k} \end{aligned}$$

The example of division of heart, which operated on the transverse slice with index $i = 490$, is presented in figure 4.6. Two yellow lines represent the left and right limits, and two green lines represent the front and back limits. Two-line segments with equations shown above cross the midpoint determined in the last step and divide the heart into four parts. The upper part is ¹ the right atrium, the left part is the right ventricle, the lower part is the left ventricle, ²² and the right part is the left atrium. The ¹³ model of heart is more realistically divided into four major components.

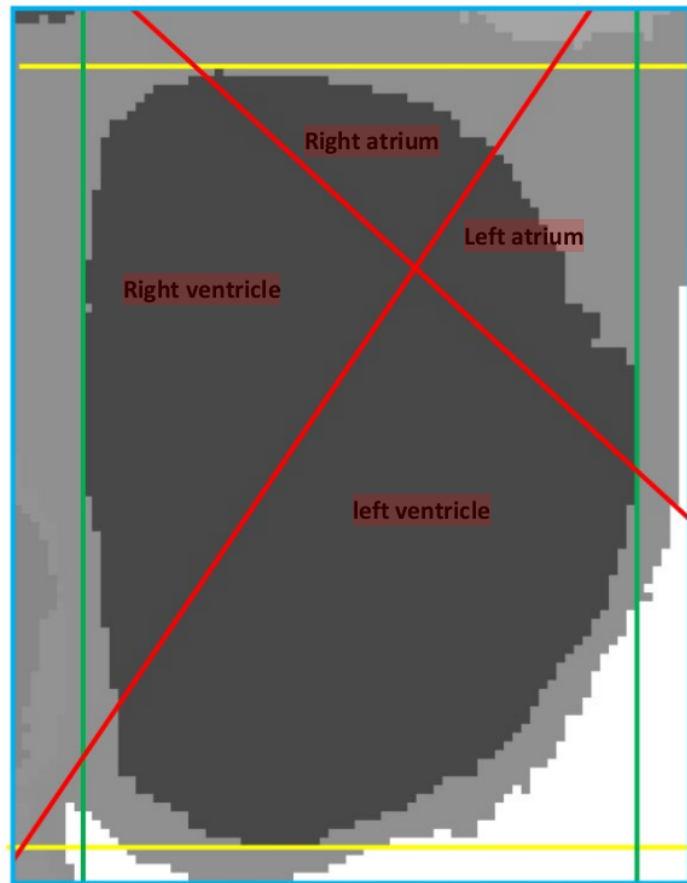


Fig. 4.6 Zoom in of the original transverse plan with index $i = 490$

4.4 Algorithm of compressing heart

The ventricle systole process is implemented by the compressing algorithm, which is introduced in this section, while the diastole process is implemented in the same way reversely. The atrium systole and diastole process are implemented with different parameters while the main algorithm is the same as the ventricle systole process. The algorithm for compressing heart is presented in two block diagrams. The calculation of the changed boundary of the heart according to the user input cardiac stage is shown in figure 4.7.

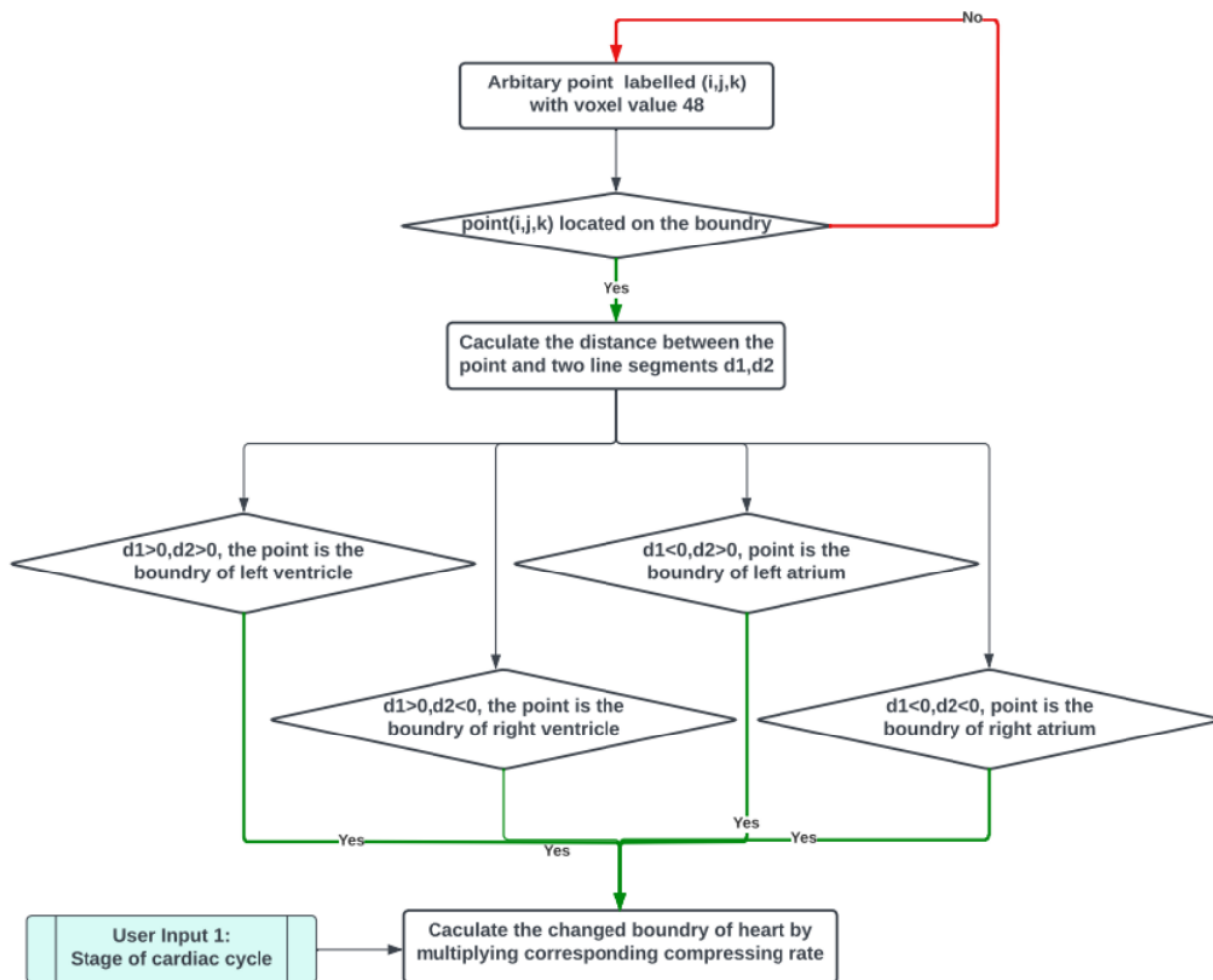


Fig. 4.7 Flowchart summarising the process of finding the changed heart boundary

The cardiac cycle is assumed to be 0.8 seconds, which is the cardiac cycle time for an adult with an average heart rate of 75 times per minute. This cycle is divided into eight time periods, the user can choose from 0 to 8 corresponding to 0 milliseconds to 800 milliseconds. The input of 0 represents the stage of the heart that is not compressed while the input of 4 represents the completely compressed stage. Inputs 1, 2, 3 are processes of systole while 5, 6 and 7 are the process of diastole. The maximum compressing proportion of left atrium boundary and right atrium boundary is 10%. The maximum compressing proportion of left ventricle boundary, right ventricle boundary is 25%. The heart is assumed to compress and relax at a constant rate, the compressing rate for calculating the changed boundary is implemented by multiplying the stage of cardiac cycle and the maximum compressing proportion. Then, the heart movement process is stimulated by changing the voxel value of those voxels outside the changed boundary from 48 to 24 or inversely.

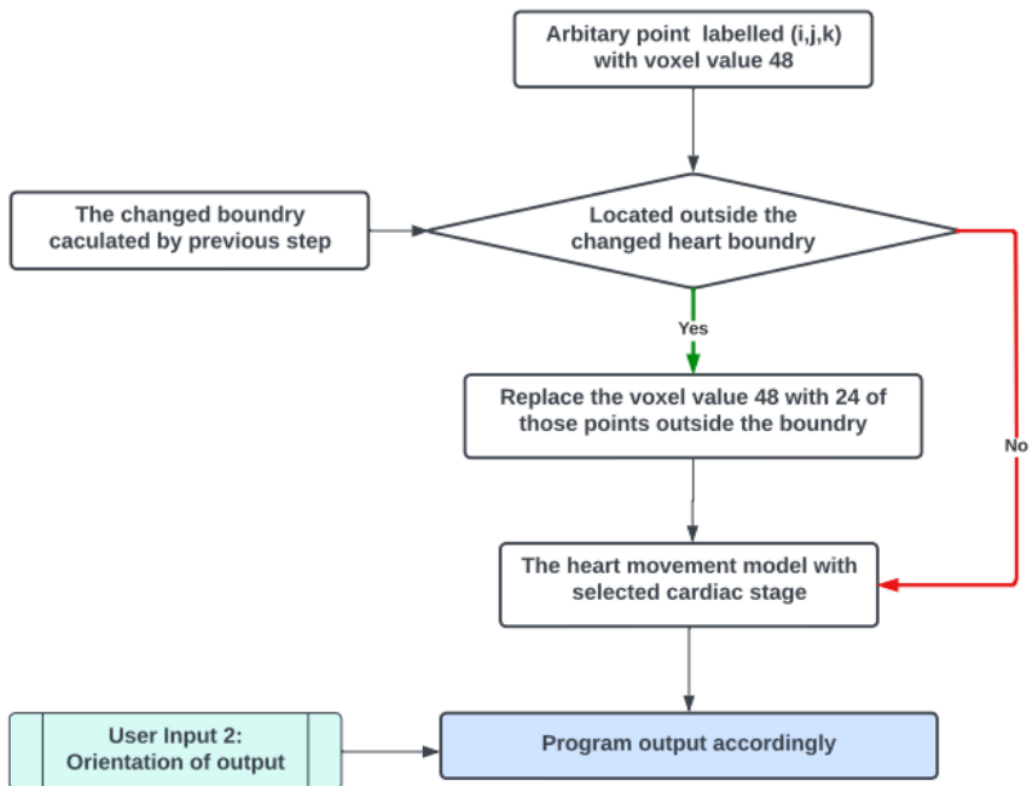


Fig. 4.8 Flowchart summarising the process of heart compressing process

4.5 User Interface

```
-----  
--- Heart Movement GUI ---  
-----  
  
Please input your desired stage of the Cardiac Cycle ( 0 (0msec) - 8 (800msec))  
4  
enter the number to choose output images  
1, output modify ventricle z-axis(transverse) view  
2, output modify ventricle y-axis(sagittal) view  
3, output modify ventricle x-axis(coronal) view  
4, output modify atrium z-axis(transverse) view  
5, output modify atrium y-axis(sagittal) view  
6, output modify atrium x-axis(coronal) view  
7, output raw data x and y-axis view  
8, ESC
```

Fig. 4.9 The user interface of the heart movement program in VSCODE terminal.

The program is written in VSCODE for Windows and compiled with the mingw-64 compiler. Two commands are to be inputted by the user. As shown in Figure 4.9, the simplified user interface requires the user to type in two commands to generate output accordingly.

- Select option 1:
- Enter stage of cardiac cycle: 0 - 8 representing 0-800 mini seconds of a cardiac cycle.
- Select option 2:
- Enter the output view 1~7 and 8 for exit.

1, Output the modified heart model in ventricle z-axis(transverse) view

2, Output modified heart model in ventricle y-axis(sagittal) view

3, Output modified heart model in ventricle x-axis(coronal) view

4, Output modified heart model in atrium z-axis(transverse) view

5, Output modified heart model in atrium y-axis(sagittal) view

6, Output modified heart model in ¹atrium x-axis(coronal) view

7, Output the raw data x and y-axis view

8, ESC

5 Results and discussion

5.1 The division of heart

The realistic of heart movement model based on the accuracy of the process of dividing heart. The algorithm of dividing heart is implemented on the transverse plane. Firstly, the circumscribed rectangle of the heart boundary is found, and the midpoint of the slice is determined. Then, two-line segments are used to segment each transverse slice into ³⁰four parts, namely left ventricle, right ventricle, left atrium and right atrium. For example, the transverse slice with index $i = 490$ is operated as shown in Figure 5.1.

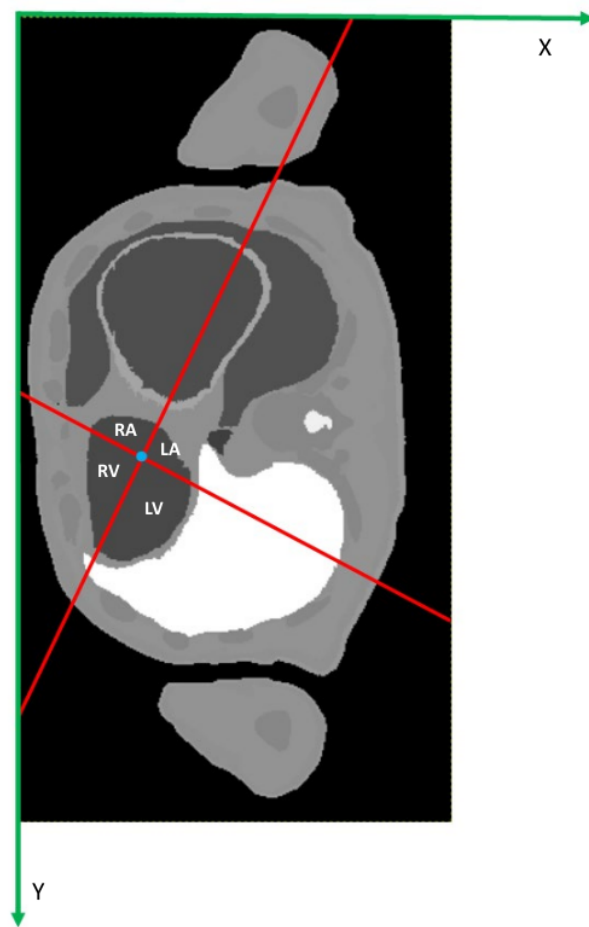


Fig. 5.1 Transverse view with index $i = 490$ divided into four parts.

The transverse slice consists of 265×490 pixels, in this slice, the heart contains 4342 pixels. The midpoint of the heart locates at $x=104.43$, $y=-273.17$, which refers to the pixel with index $(i, j, k) = (490, 105, 274)$. Two-line segments with the equations: $y = 1.743 * x - 440.34$ and $y = 0.569 * x - 199.68$ divide the heart into four parts as shown in figure 5.1. RA refers to right atrium, which contains 668 pixels. RV refers to right ventricle, which contains 1336 pixels. LA refers to left atrium, which contains 501 pixels. LV refers to left ventricle, which contains 1837 pixels.

According to the research made by Lucy E. Hudsmith, for normal human beings, the volume and size of (LA)left atrium, (RA)right atrium (LV)left ventricle and (RV)right ventricle have an average range. LV and AV volumes and LV mass were larger in males than females: LV end-diastolic volume 160 ± 29 mL vs. 135 ± 26 mL, LV end-systolic volume 50 ± 16 mL vs. 42 ± 12 mL; RV end-diastolic volume 190 ± 33 mL vs. 148 ± 35 mL, RV end-systolic volume 78 ± 20 mL vs. 56 ± 18 mL.[18] Compare the number of pixels obtained by previous step and the statistical results, the ratio is as expected. An MR image of transverse section of heart is shown in Figure 5.2, which is similar to the division of heart as shown in figure 5.1. The algorithm of dividing heart is proven to be reasonable.

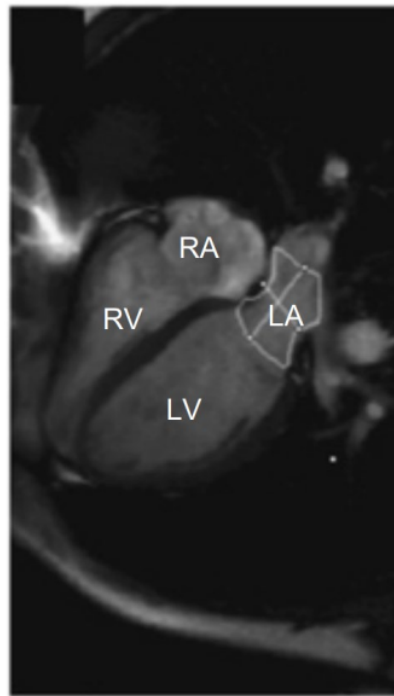


Fig. 5.2 The transverse section of heart obtained by MRI [18]

5.2 The simulation of cardiac cycle

The simulation of cardiac cycle is implemented by compressing the heart to different degree according to the selected cardiac stage. User input 0 for the stage of ending of diastole, input 4 for the stage of ending of systole. Input 1,2 and 3 are the process of systole while 4,5 and 6 are the process of diastole. The program can generate output according to the orientation of the heart model that user selects. Three featured slices of the output files are selected as an illustration in this section for transverse view, coronal view and sagittal view.

5.2.1 The simulation of a complete cardiac cycle (in coronal view)

A complete cardiac cycle is presented from ¹⁴figure 5.3 to figure 5.11. The figure 5.3 shows the shape heart at the start of a cardiac cycle. The systole period is represented by figure 5.4 to figure 5.6, during which the heart keeps compressing. The figure 5.7 shows the shape of heart at the ending of systole when it is completely compressed. The diastole period is represented by the process from ¹⁴figure 5.8 to figure 5.10. The figure 5.11 shows the shape of heart at the ending of diastole when it is at complete relaxation.



Fig. 5.3 The coronal view of heart at 0ms of cardiac cycle



Fig. 5.4 The coronal view of heart at 100ms of cardiac cycle



Fig. 5.5 The coronal view of heart at 200ms of cardiac cycle



25 Fig. 5.6 The coronal view of heart at 300ms of cardiac cycle



25 Fig. 5.7 The coronal view of heart at 400ms of cardiac cycle



Fig. 5.8 The coronal view of heart at 500ms of cardiac cycle



Fig. 5.9 The coronal view of heart at 600ms of cardiac cycle



Fig. 5.10 The coronal view of heart at 700ms of cardiac cycle



Fig. 5.11 The coronal view of heart at 800ms of cardiac cycle

5.2.2 End of systole VS end of diastole (In transverse view)

A comparison of the ventricle shape when it is completely compressed at the ending of systole and when it is at complete relaxation at the ending of diastole is made in transverse view. As shown in Figure 5.12. As shown in figure 5.12, the index of the selected transverse slice as an illustration is $i = 490$. The left sub-image is the heart at the end of diastole, the right sub-image is the heart at the end of systole. The middle sub image highlights the changing range of heart during the process cardiac cycle. The figure 5.13 shows the estimation of changing range of heart ventricle in cardiac cycle, which is as expected from previous research part.

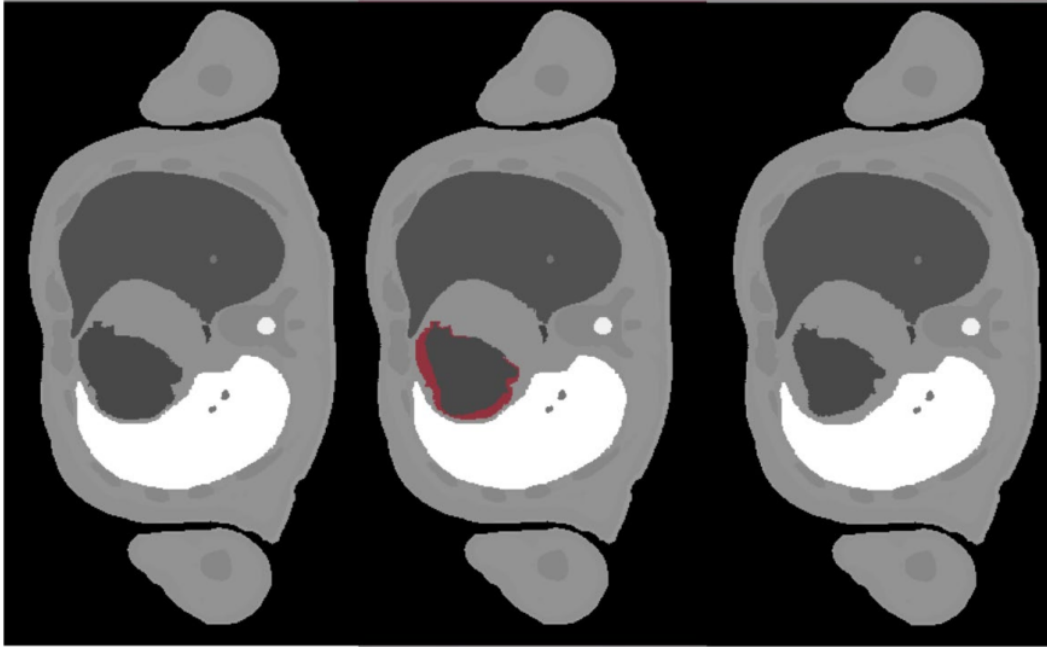


Fig. 5.12 A comparison of the shape of ventricles at the end of systole and the end of diastole in transverse slice with index $i = 490$.

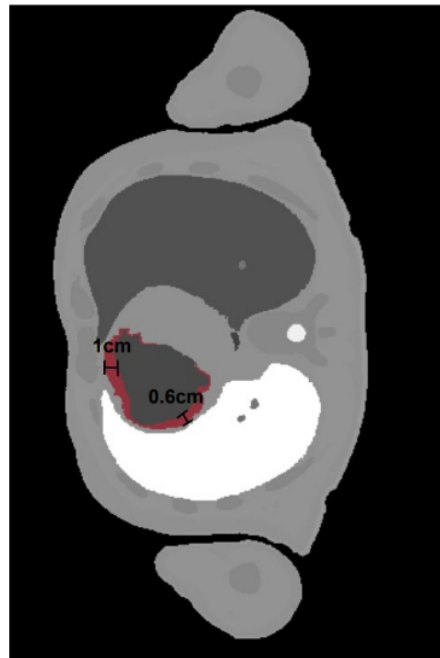


Fig. 5.13 Estimation of deformation size of ventricles in transverse slice with index $i = 490$.

6 Conclusions and future work

6.1 Achievements

In this project, the simulation of heart movement in a cardiac cycle is developed. The heart model is made to be more realistic to the real heart by dividing ⁴¹the heart accordingly. The heart is divided into four parts by two-line segments in each transverse slice, the process of which is proven to be reasonable. The systole period and diastole period of cardiac cycle are implemented by the compressing algorithm. Ventricles and atriums are compressing with corresponding changing rates, which makes the heart movement model more realistic. The team behind this project guided by Dr. Fumie is developing a device which uses microwaves to detect dyspnea. This project contributes to the development of DHP, which is used to the device, by adding a more realistic heart movement model to the original static DHP.

6.2 Limitations

The PGM files are used to store the static DHP which obtained by MRI scanning on real people. The DHP data is loaded in a 3D array in the Fortran program. It is not convenient or straightforward to check and compare the DHP in different views and stages of cardiac cycle. Some featured slices are selected, but not clear or intuitive. The author tried to use mesh plot in MATLAB to visualize the 3D model, however, the computation burden is too heavy for personal computer.

According to the previous literature research part, the heart size and shape of male and female can be significantly different. The heart rate for normal people and people with dyspnea can also be different. The average statistic data for normal people utilized in this project are supposed to be adapted corresponding to specific people.

6.3 Future plans

Currently, the compressing algorithm is implemented by replacing voxel value 48(heart) with 24(fat). It should be noted that the heart is surrounded by two lungs, some areas should be replaced by other surrounding organs and tissues. It is a necessary step for the combination of heart movement in a cardiac cycle and lung movement in a respiration cycle. Only when this problem is solved can the DHP be utilized to test the device which uses microsound to detect dyspnoea.

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